

Bengali Hateful Memes Detection: A Comprehensive Dataset and Deep Learning Approach

Abstract—The rise of hateful memes promoting racism, misogyny, and discrimination on social media necessitates advanced detection methods to counter their increasing prevalence. These memes, often distributed widely through social platforms, have the potential to normalize prejudice and influence public opinions negatively. In response to the growing meme culture, this research addresses the need for enhanced automatic detection of hateful content. Employing five pre-trained deep learning models (VGG16, ResNet50, InceptionV3, DenseNet 21, and Xception), we conducted experiments to classify hateful and non-hateful memes. Our Bengali dataset, comprising 3000 memes, was utilized for the study. Results indicate that Xception exhibited superior performance with an accuracy of 68.35%. This research aims to contribute to the development of effective tools for combating the spread of harmful content on social media platforms.

I. INTRODUCTION

In the age of digital communication and social media, memes have become an integral part of online culture. Memes, often characterized as humorous or sarcastic images or videos, possess a remarkable ability to convey ideas, emotions, and opinions in a straightforward and engaging format. However, this widespread phenomenon has also seen the emergence of hateful memes, which propagate degrading, discriminatory, or offensive content targeting specific individuals, groups, or communities [1].

The majority of hateful memes pose a significant challenge in maintaining a safe and inclusive online environment. Detecting and mitigating such content is crucial to restraining the spread of hate speech, fostering online civility, and protecting individuals from cyberbullying and discrimination. While substantial efforts have been made in detecting and categorizing harmful content in major languages, there remains a notable gap in addressing this issue within the Bengali language [2], which boasts a vibrant online community. The lack of NLP tools, academic resources, and large labeled Bengali meme datasets poses challenges in understanding memes' hidden meanings.

For instance, in figure 1a and 1c, these memes may appear amusing at first sight. However, upon closer look, it becomes clear that they are actually insulting and disrespectful. On the other hand, 1b is a humorous meme that does not contain any hateful content. Categorizing memes as either hateful or non-hateful, and creating sophisticated multimodal Bengali meme analysis systems is a difficult task. Furthermore, the analysis of hateful memes in Bengali can require the integration of cross-modal knowledge and comprehension of multimedia content.

The need for hateful meme detection in Bengali arises from the exponential growth of Bengali-speaking internet users and the consequent rise in content generation. Existing models and datasets may not accurately capture the Bengali language hate memes. Moreover, a publicly available Bengali dataset will help to encourage research in this area. However, there exists very little work on Bengali hate meme detection with properly available datasets. [3]

In this paper, we address the critical gap in hateful meme detection for Bengali language by proposing a Deep Learning-based classification model. Our approach utilizes an ensemble of deep learning techniques, including VGG16, ResNet50, InceptionV3, and DenseNet121, to recognize and differentiate between Hateful and Non-hateful memes.

The core contribution of our research lies in the development of a customized dataset comprising 3000 Bengali hateful memes. Additionally, we present a methodology that leverages the distinct layers of the CNN, particularly emphasizing the convolution layer, to effectively discern and classify hateful content. Experiment results demonstrate the effectiveness of our proposed model in improving accuracy and reliability in detecting hateful memes in the Bengali language. Providing the dataset for research



Figure 1: Example of meme's multimodal nature

and development allows the community to build upon our initial efforts in this under-explored domain and language. By making the dataset available to the growing research community, we can catalyze future research that improves the accuracy of detecting and categorizing internet memes.

II. RELATED WORK

Recent research has focused on detecting hate speech in memes through unimodal and multimodal approaches. While unimodal methods analyze only textual or visual aspects, multimodal techniques combine both to understand memes better. However, there is a lack of publicly available Bengali meme datasets. This review covers relevant literature on hate speech detection in Bengali and English, as well as multimodal meme analysis. It highlights the need for publicly shared Bengali meme datasets and our contributions towards multimodal hate speech categorization in Bengali memes.

Hossain et al. (2022) [3] proposed a Bengali dataset of 4158 memes, providing a detailed annotation guideline, conducting extensive experiments with visual and textual models, and integrating features from both modalities using the early fusion approach. This paper showed significant improvement in hateful meme classification through joint evaluation of visual and textual features, and the effectiveness of multimodal features compared to unimodal features for meme classification. However, the availability of the dataset was not stated.

Ahmed et al. (2023) [4] demonstrated effective detection of cyberbullying in Bengali memes, addressing the growing need for computational tools to identify bullying behavior online. They developed a unified deep neural architecture for classifying memes gathered from various social media platforms as bullying or non-bullying with an accuracy of 87% and an F1 score of 88%.

The paper by Kiela et al. (2020) [5] introduces a new challenge set for multimodal classification, provides baseline performance numbers for unimodal and multimodal models, and highlights the potential for progress in multimodal research and addressing real-world problems. The dataset used in the study is the "Hateful Memes dataset" which consists of 10,000 multimodal memes for detecting hate speech. [6]

In [7], Sharman et al. (2021), they proposed two novel problem formulations: detecting harmful memes and the social entities that these harmful memes target. They also released a new benchmark dataset, HarMeme, developed based on comprehensive annotation guidelines. The dataset contains 3,544 memes related to COVID-19. The dataset was used for benchmarking state-of-the-art unimodal and multimodal models for detecting harmful memes and identifying their targets. The dataset is distributed for research purposes only, without a license for commercial use.

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Sharma et al. (2020) [8] emphasized the need for research in the automatic processing of Internet memes, the success of the challenge with a large number of participants and submissions, and the reported F1 scores for sentiment and emotion classification tasks. They used the dataset given by FacebookAI [6]. They had some limitations including the scarcity of properly annotated meme data, the need for enrichment of the dataset with annotations for different language memes, the motivation to organize similar events in the future, and the need for further research and improvement in the field of meme emotion analysis.

Karim et al. (2022) [9] conducted a study on detecting hate speech in memes in the Bengali language. They highlighted the challenges of interpreting localized cultural references in multimodal content. To address this issue, they proposed a novel approach that accurately identifies hate speech in Bengali memes and texts. As part of their study, they created a large dataset for hate speech detection in Bengali using multimodal techniques. They employed early-and late-fusion techniques, transformer-based neural architectures, and CNN architectures for analyzing the text and imaging modalities.

The paper by Chi et al. (2017) [10] proposes the use of simple models to combine information from various sources and provide robustness to different types of input. Their study has achieved high accuracy in the emotion classification domain. They have used a dataset that includes data from textual and visual modalities, including images labeled into 8 emotion classes obtained from the Flickr dataset and submissions from Reddit containing images.

We have developed a new approach for detecting hateful memes, which involves binary classification using only visual information. Our method uses deep convolutional neural networks (CNNs) such as VGG16, ResNet50, DenseNet121, InceptionV3, and Xception models, pre-trained on ImageNet photos, and then customized to classify memes as hateful or non-hateful. Our Bengali meme dataset consists of 3000 hand-labeled images gathered from various social media platforms. Our study is the first to explore visual-only models for binary hate speech detection with Bengali memes. Earlier approaches to meme analysis relied heavily on both visual and textual features. Our image-focused methodology aims to establish an initial benchmark for future systems using both image and text. In the future, we plan to combine textual

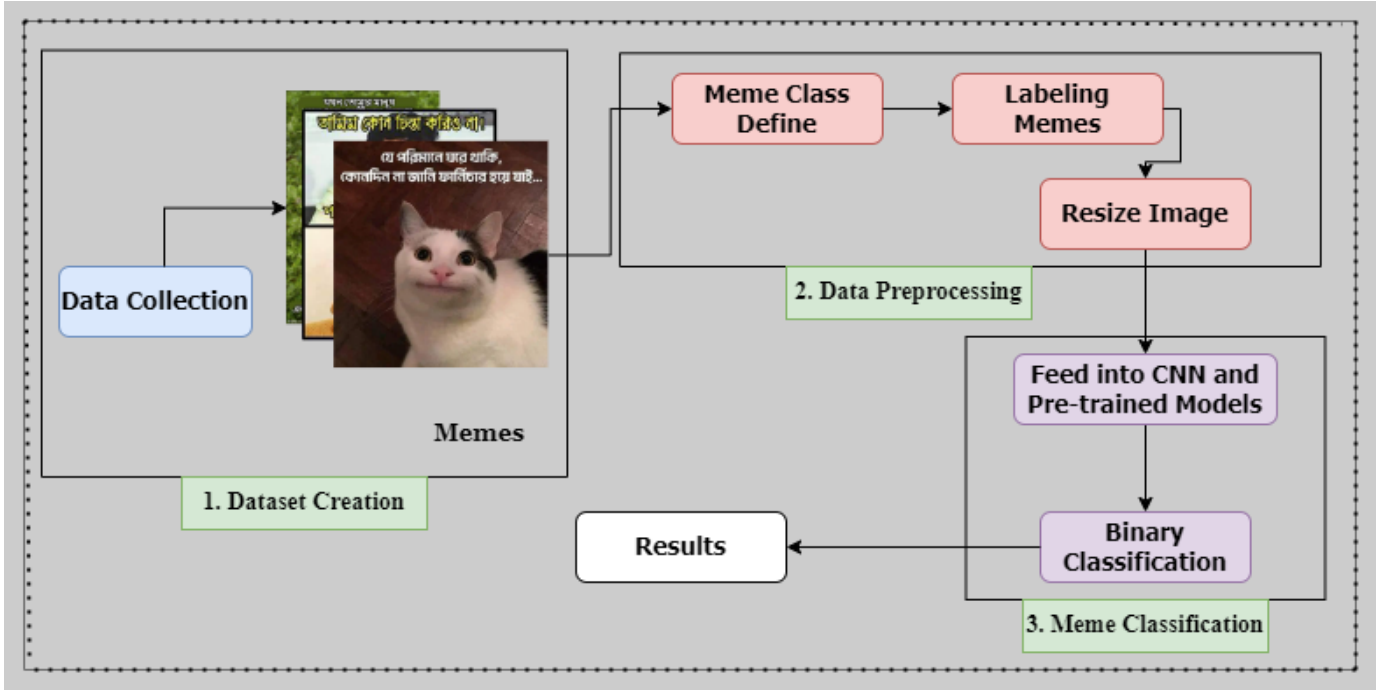


Figure 2: Overall Workflow

models with images to improve the accuracy of our model.

III. PROPOSED METHODOLOGY

The proposed methodology of Bengali hate meme classification is shown in Figure 2. There are three main phases in our classification approach, dataset creation, data preprocessing, and meme classification. We will examine each part closely in this section to explain the techniques used at every stage. The sections below will detail the overall workflow of our work.

A. Dataset Creation

As we have created our own dataset of 3000 Bengali memes¹, we had to collect the memes manually from Facebook, Instagram, Reddit, and X(Previously known as Twitter).

As previously stated, the curated dataset encompassed 3000 memes equally balanced between 1500 hateful and 1500 non-hateful samples. Maintaining class equilibrium helps mitigate overfitting towards any specific category, unlike imbalanced distributions where particular classes disproportionately influence model convergence. Initial parity preserves model generalization and avoids degraded performance in minority classes. Beyond the base training data, we have collected an additional of approximately 750 memes that will be further added to the current dataset by adding more samples during subsequent expansion.

At first, we collected posts and tweets by searching for relevant tags. Specifically, meme contents were gathered by querying tags including "meme", "Bangla hate

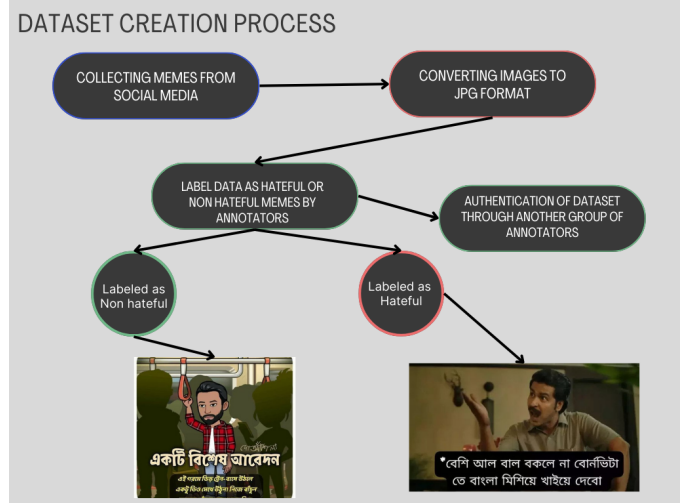


Figure 3: Dataset Creation Process

memes", "Bangla meme", "funny", and "Bangla jokes". Also, there were some Bangla memes pages on Facebook, which helped us to construct the dataset. This tagging functionality allowed us to filter for social media items that are more likely to contain comedic intent or content. Figure 3 visualizes the process.

B. Data Preprocessing

Before feeding the dataset into the networks, preprocessing is conducted. We resized the image into the size 224 X 224 X 3, which made the images suitable to drive

¹<https://shorturl.at/dmtCL>

into the CNN models. The following figure demonstrates the overall training and testing phases.

We have currently labeled our dataset using two classes only: hateful and non-hateful memes. This binary classification format allows initial testing of how visual-only models perform with our data. Generally, a meme is defined as hateful if it propagates any type of negativity, insult, or harm targeting other people groups. Whereas non-hateful or benign memes aim to express sarcasm or humor positively without malicious intent.

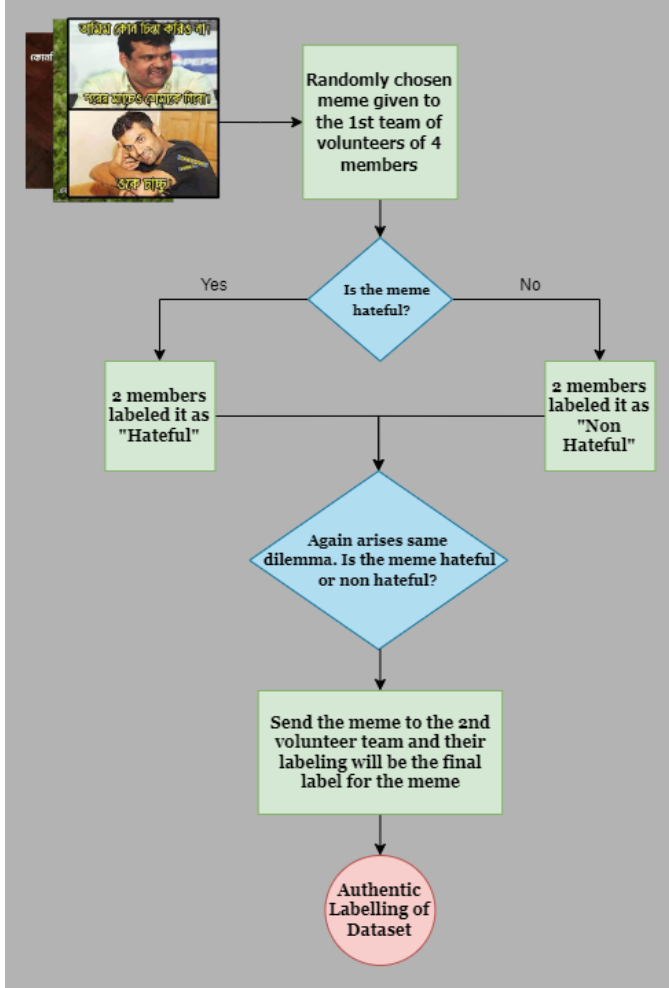


Figure 4: Labeling Process of Memes

Figure 4 illustrates the labeling process of the memes. To ensure unbiased labeling, the annotator teams were kept independent without knowledge of our research objectives. Initially, each member labeled the memes individually without internal consultation. This enabled them to provide instinctive judgments without group conditioning. After solitary labeling completion, the teams conferred to resolve any disagreements based on the provided characteristics guide, referencing solely textual and visual signs. Memes lacking agreement were excluded from the final dataset, upholding label integrity through rigorous

vetting. This structured 3-phase procedure - isolated tagging by individuals, collaborative conflict mediation, and guidelines-dependency verification - safeguarded the reliability of ground truth labels across hateful and non-hateful categories.

For some memes, the first team of annotators was unable to reach an agreement on whether a meme should be labeled as hateful or non-hateful. In that case, the meme was labeled according to the majority vote. However, in cases where the votes were tied 2-2 between hateful and non-hateful, the meme was sent to a second team of annotators for further review to ensure data accuracy. Approximately 700 memes fell into this tied category which created confusion for the initial group of annotators and required additional authentication from the second team.

C. Meme Classification

Since memes comprise image and text modality, we focused first on image modality using convolutional neural networks (CNNs) - the state-of-the-art for this task. Our paper explores CNN structure [11] and function in computer vision applications for visual modalities.

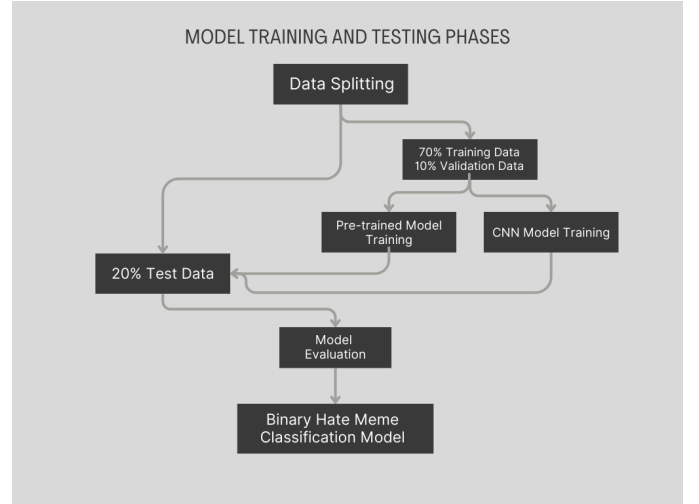


Figure 5: Training and Testing Phases

Figure 5 depicts the data splitting and model train and testing phases. 70% data was kept for training where 10% was for validation. Rest of the 20% is for model evaluation process.

Five CNN architectures are employed including VGG16 [12], ResNet50[13], InceptionV3[14], DenseNet121[15], and Xception[16]. The base models are initialized with pre-trained weights and their layers are frozen for transfer learning, leveraging their learned feature extraction capabilities. All the models have a dense output layer with a sigmoid activation, which is added to make a binary prediction.

1) *CNN*: A three-layer convolutional neural network was developed for visual classification of memes. The input layer ingests pixel-level representations to be filtered for patterns. The first two convolutional layers apply 16 and 32 kernels of size 3x3 respectively to identify elementary visual motifs. After each convolution, 2x2 max pooling compresses the spatial encodings by halving width/height. A third convolution with 16 filters precedes global max pooling to fully condense the feature maps into vector embeddings. These embeddings feed into fully connected layers ending in a sigmoid-activated node for binary hateful vs non-hateful classification.

2) *VGG16*: A VGG16 model pre-trained on ImageNet initializes the convolutional base, with classification layers removed. The truncated VGG16 feeds into a Sequential pipeline adding flattening to unroll 3D outputs. A 512-unit dense ReLU-activated layer follows for dimensionality reduction, outputting activations to a final sigmoid layer for binary classification.

3) *ResNet50*: A sequential model is constructed with a ResNet base model as the initial layer. This is followed by flattening the features from ResNet. Two dense layers of 2048 and 512 units with ReLU activation are then added with dropout regularization of 0.5 and 0.25 respectively. Finally, a dense output layer with a single sigmoid-activated unit is added for binary classification. The model takes in embedding features as input to the ResNet base. The dense layers act as a classifier on top of the compact features extracted by ResNet through its convolutional and pooling operations. Dropout regularization is used to reduce overfitting.

4) *InceptionV3*: This model uses transfer learning with a pre-trained Inception v3 model. The Inception model provides already-learned feature extraction. Global average pooling condenses the Inception features. A dense layer with ReLU activation and dropout regularization is added. Finally, a dense layer with a sigmoid activation makes a binary classification.

5) *DenseNet121*: Like previous models, the DenseNet model is added to a sequential model. Flattening is applied to condense the features. Two dense layers with 1024 and 512 units, ReLU activation, and dropout regularization are added.

6) *Xception*: The model utilizes transfer learning with a pre-trained Xception model. The layers of Xception are set to non-trainable to leverage its pre-learned features. The Xception model is added to a Sequential model. Global average pooling is applied on top to reduce the spatial features to a fixed-length vector. Two dense layers of 1024 and 512 units are added with ReLU activation and dropout regularization of 0.5 and 0.3 respectively.

The models were trained for 20 epochs with 65 steps per epoch and a batch size of 32. We utilized the Adam [17] optimizer for stochastic optimization.

IV. EXPERIMENTAL RESULTS

This section presents a comprehensive analysis of the performance of the models in terms of precision, recall, F1-score, and accuracy. The impact of dropout layers on model performance is also discussed, along with comparisons of training and validation accuracy and loss graphs.

Table I: Model Hyperparameters

Model	Learning Rate	Dropout Values
CNN	0.0001	-
VGG16	0.0001	-
ResNet50	0.00001	0.5, 0.25
InceptionV3	0.0001	0.2
DenseNet121	0.0001	0.5, 0.3
Xception	0.0001	0.5, 0.3

Table I illustrates the model names along with their learning rate and dropout layer values. It is also noted from Table I that Resnet50 has a different learning rate compared to the others. This is because after conducting several experiments, it was observed that this particular learning rate worked better than other values.

The following Table II demonstrates the outcome of the models in terms of Precision, Recall, F1-score, and accuracy for binary hate meme classification.

Table II: Performance Comparison of our Visual Models

Models	Precision	Recall	F1-Score	Accuracy
CNN	0.6265	0.6667	0.6406	0.6392
VGG16	0.6617	0.6840	0.6595	0.6566
ResNet50	0.6090	0.7134	0.6534	0.6218
InceptionV3	0.6667	0.6051	0.6240	0.6535
DenseNet121	0.6947	0.5871	0.6338	0.6709
Xception	0.6571	0.7419	0.6874	0.6835
ResNet50 without Dropout				
ResNet50	0.7439	0.3697	0.4864	0.6044

It is seen that Xception has the highest recall which is 74.19% and F1-score of 68.74%, while the ResNet50 model without a dropout layer has the highest precision of 74.39%. However ResNet50 without dropout layer has not been taken into consideration as the overall accuracy falls.

From Figure 6 it is seen that, without a dropout layer, the train-validation accuracy and loss graph give spikes, which is unusual. After conducting several experiments, we observed that the addition of dropout layers significantly enhances the accuracy of every model, except for VGG16. This improvement is shown in Table II. The accuracy varies across the models, with Xception having the highest accuracy of 68.35% among all models.

The ResNet50 model without dropout layers exhibits a decrease in overall performance, as indicated in Table II. This contrast is illustrated in the accompanying Figure 6, displaying the graphs of ResNet50 models with and without dropout layers. The orange line represents validation, and the blue line represents training. Dropout layers were

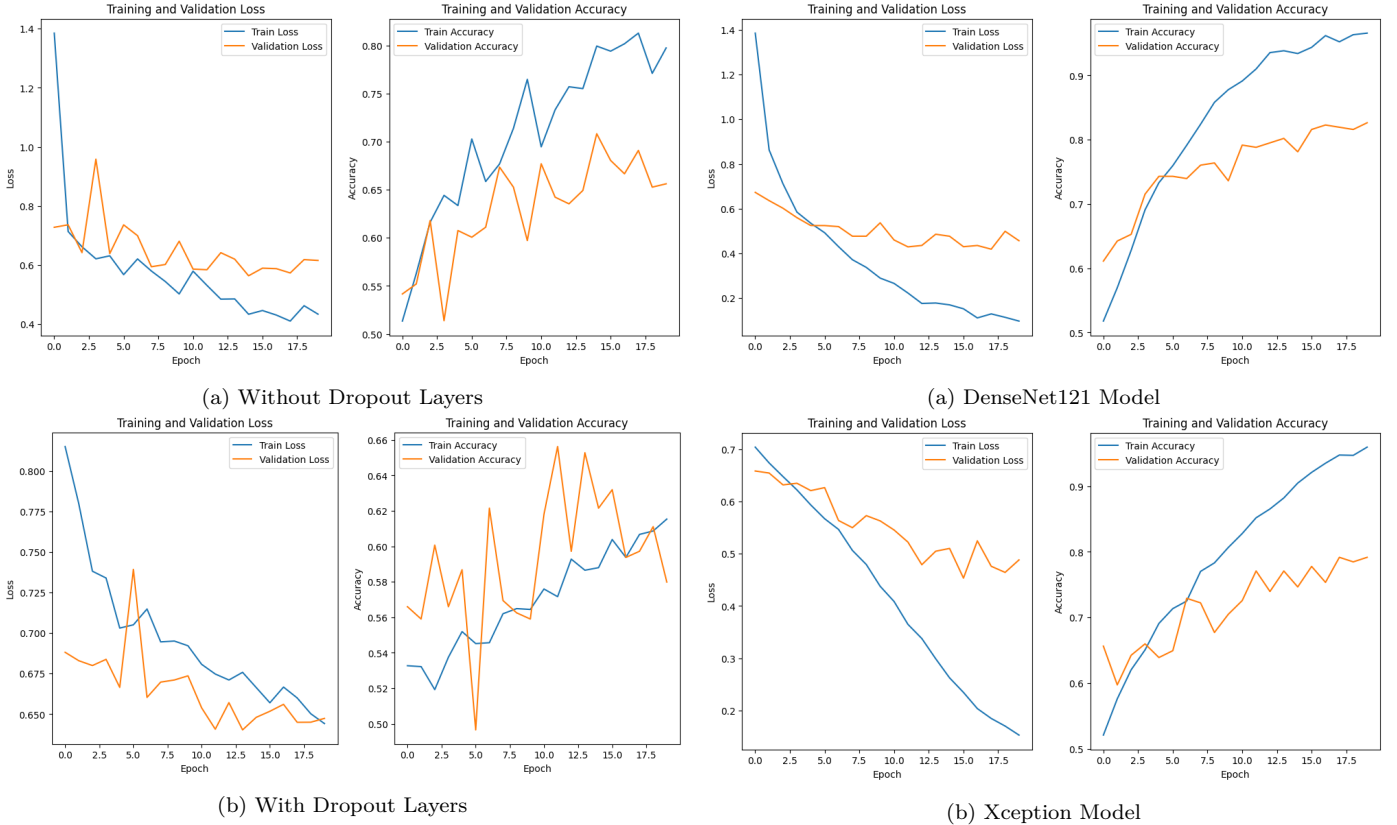


Figure 6: Comparison of Training and Validation Acc/Loss Graphs of ResNet50 Model

incorporated into ResNet50, InceptionV3, DenseNet121, and Xception to enhance the overall accuracy.

Figure 7 illustrates the training loss, validation loss, training accuracy, and validation accuracy of our DenseNet, Xception, and Inception models. Notably, the DenseNet and Xception models achieved a training accuracy of approximately 98% and a validation accuracy close to 80%, surpassing the performance of all other models. The Inception model also demonstrated commendable results, with a training accuracy of nearly 99.00% and a validation accuracy of around 80%. The loss graph for each model represents the loss after 20 epochs.

Having employed various models for the classification of memes solely based on images, we compared our accuracy results with those of other research works [5] [7] [18] in a similar domain. It was observed that the accuracy ranges in most cases fell between 50%-65%. Despite the application of different models by these studies, we conducted this comparison as a unimodal approach.

V. FUTURE WORK

There remain several promising directions to explore in advancing hate speech detection for Bengali memes. Our future work will focus on three primary areas: gathering additional Bengali memes to expand the current dataset,



Figure 7: Training and Validation Accuracy/Loss Graphs of DenseNet, Xception and Inception Model

developing enhanced multimodal classification architectures to leverage both image and text and experimenting with ensemble methodologies by combining multiple base models.

Firstly, we will expand our Bengali meme image dataset to over 6,000 samples. A larger variety exposes more ethnicities, dialects, and niche memes. Secondly, we will use dual-input neural architectures, which will fuse visual and textual streams to enhance inter-modality robustness, as cross-modal models like ViLBERT align latent signals successfully. Finally, using ensemble learning will combine inferences from visual, linguistic, and lexicon-based

classifiers to mitigate individual weaknesses. We believe pursuing expanded data, multimodal integration, and model blending remains integral for generalized, robust hate meme detection.

VI. CONCLUSION

In this research, we compiled a dataset of Bengali internet memes to support and advance research in this field. Utilizing a unimodal approach through binary classification, we analyzed solely visual information for the identification of hateful memes online. Our highest-performing model attained an accuracy of 68.35% in distinguishing between hateful and non-hateful memes.

However, our goal is to further enhance prediction accuracy by categorizing hateful memes into more specific subclasses based on the target of hate (e.g. religion, political parties, celebrities, feminism, sports, etc.). To achieve this, future work will involve incorporating multimodal detection by integrating textual analysis using models such as BERT along with image classification. By comprehensively considering both textual and visual modalities, we aim to develop a robust multimodal approach for improved hate speech detection in memes.

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